Astro HW 10

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NOTE: Work on the page itself is kinda sparse, but all the code used for the exact results is at the end[1]

10.1 Supernova! Wahoo! Yipee!

a) Tme since SN

Given ...

$$\theta = \frac{D}{d}$$

We find ...

$$D_{snr} = (\theta_{snr} \times d_{snr}), \qquad R_{snr} = \frac{D_{snr}}{2}$$

$$t_0 = R_{snr}/V_{snr}$$

$$\boxed{t_0 \approx 3.269 \times 10^9 \text{s} \approx 1.03 \times 10^2 \text{yr}}$$

b) Neutrinos per sec

Given...

$$F = \frac{L}{4\pi d^2}$$

We find that

$$F = 1.156 \times 10^{16} \text{m}^2$$

$$F_t = \frac{F}{\Delta T}$$

$$\boxed{F_t \approx 7.712 \times 10^{14} \text{ Neutrinos/s}^2}$$

[1] pls dont dock me for not showing enough work ;-;

c) Magnitudes and extinction

Given ...

$$M_{bol} = M_{bol\odot} - 2.5\log(L_{bol}/L_{\odot}), \qquad M_{bol\odot} = 4.74 \qquad L_{bol} = 10^9 L_{\odot}$$

$$m = M + 5\log(d) - 5, \qquad d = 8.5\text{kpc}$$

$$m_{obsc} = M + 5\log(d) - 5 + A_v \qquad A_v = 30$$

We find ...

$$M_{bol} = 4.474 - 2.5\log(10^9 L_{\odot}) \approx -17.76$$

$$m_{bol} = -18.1 + 5\log(8500\text{pc}) - 5 \approx -3.113$$

$$m_{Obscbol} = -18.1 + 5\log(8500\text{pc}) - 5 + 30 \approx 26.886$$

$$\boxed{M_{bol} \approx -17.76, \qquad m_{Obscbol} \approx 26.887}$$

This means it is completely impossible to resolve with the naked eye, and, while resolvable with some scopes, still quite difficult.

10.2 Stellar remnants and angular momentum

Given ...

$$I = \frac{2}{5}MR^2$$

$$L = I\omega$$

$$\omega = \frac{\theta}{t}$$

$$p_{sun} = 25.4\text{days}, \qquad r_{wd} = 6000\text{km}$$

We find ...

$$I_{sun} = 2/5 m_{\odot} r_{\odot}^2$$

$$\omega_{sun} = (2\pi)/p_{sun}$$

$$L_{sun} = I_{sun}\omega_{sun} \approx 1.1 \times 10^{42}\text{m}^2\text{kg/s}$$

$$I_{wdSun} = 2/5 m_{\odot} r_{wd}^2$$

By conservation of angular momentum...

$$\omega_{wdSun} = L/I_{wdSun}$$

$$P_{wdSun} = (2\pi/\omega_{wdSun})$$

$$\boxed{P_{wdSun} \approx 163.2\text{s}}$$

10.3 The nature of pulsars

a) Orbiting WDs

Given...

$$p^2 = \frac{4\pi^2}{G(m_1 + m_2)}$$

$$M_{wd} = .7M_\odot, \qquad r_{wd} = 7 \times 10^6 \text{m}, \qquad P_{wd} = 1.337\text{s}$$

We find that ...

$$a = \left(\frac{(p_{wd}^2)(2GM_{wd})}{r\pi^2} \right)^{1/3}$$

$$\boxed{a \approx 2 \times 10^6 \text{m}}$$

b) Is this possible?

No, this is not possible. This is because $\boxed{a < r_{wd}}$, meaning that the WDs semimajor axis would be literally within each other, and thus they would have to be merged

c) WD Period

Given ...

$$v_{rot} = \frac{2\pi R}{P} \rightarrow P = \frac{2\pi R}{v_{rot}}$$

$$v_{rot}(P, M) \approx .1c \left(\frac{P}{1s} \right)^{-1} \left(\frac{M}{.7M_\odot} \right)^{-1/3}$$

We find that ...

$$v_{rot} \approx 2.662 \times 10^7 \text{m/s}$$

$$P = 2\pi \frac{r_{wd}}{v_{rot}}$$

$$\boxed{P \approx 1.652\text{s}}$$

d) Breakup Speed

Given ...

$$a_c = \frac{v^2}{r}$$

$$g = \frac{GM}{R^2}$$

We find ...

$$v_{crit} \rightarrow \frac{v^2}{r} = \frac{GM}{R^2}$$

$$\frac{v^2}{\cancel{r}} = \frac{GM}{R^{\cancel{2}}}$$

$$\boxed{v = \sqrt{\frac{GM}{R}}}$$

e) Breakup speed from alternative function

Given...

$$v_{crit} = \sqrt{\frac{GM_{wd}}{r_{wd}}}$$

We find that ...

$$\boxed{v_{crit} \approx 3.643 \times 10^6 \mathrm{m/s}}, \qquad \boxed{v_{crit} \approx 3.583 \times 10^6 \mathrm{m/s}}$$

f) Is this feasible?

No shot... This is because $v_{rot} > v_{crit}$, meaning that a WD rotating that fast would break up.

g) What if it was a NS?

Given ...

$$P = 2\pi \frac{r_{NS}}{v_{rot}}$$

$$v_{crit} = \sqrt{\frac{GM_{NS}}{r_{NS}}}$$

$$m = 1.4M_{\odot}, \qquad r_{NS} = 10\mathrm{km}$$

We find...

$$v_{rot} \approx 4.7 \times 10^4 \mathrm{m/s}$$

$$v_{crit} \approx 1.36 \times 10^8 \mathrm{m/s}$$

This means that a NS spinning with that rate *does* make sense, this is because $v_{rot} < v_{crit}$, meaning that the NS will *not* break apart, and could have a pulse of 1.337s

10.4 Mass models of the Milky Way Galaxy

Given ...

$$v**2 = \frac{GM}{R} \to m = \frac{v^2 r}{G}$$

We find ...

$$m_0 = \frac{v_0^2 r_0}{G}, \qquad m_1 = \frac{v_1^2 r_1}{G}, \qquad m_2 = \frac{v_2^2 r_2}{G}, \qquad m_3 = \frac{v_3^2 r_3}{G}$$

$v_c(\text{km}^{-1})$	$r(\text{kpc})$	$M(r)(kg)$
255	0.4	1.20×10^{40}
192	2.8	4.77×10^{40}
220	8	1.79×10^{41}
235	17	4.34×10^{41}

10.5 Galactic center black hole and S0-2

a)

Given...

$$M_{bh} = \frac{4\pi^2 a^3}{Gp^2}$$

We find ...

$$M_{bh} \approx 7.22914 \times 10^{36}\text{kg}$$

b)

Given...

$$r = a(1 \pm e)$$

$$v_a = \left(\frac{GM_{bh}}{a}\frac{1-e}{1+e}\right)^{1/2}$$

$$v_p = \left(\frac{GM_{bh}}{a}\frac{1+e}{1-e}\right)^{1/2}$$

$$a = 960\text{AU}, \qquad M_{bh} = 7.22914 \times 10^{36}\text{kg}$$

We find

$$\boxed{r_a \approx 2.618 \times 10^{14}\text{m}}, \qquad \boxed{r_p \approx 1.91 \times 10^{13}\text{m}}$$

$$\boxed{v_a \approx 4.892 \times 10^5\text{m/s} \approx .16\%\text{speed of light}}$$

$$\boxed{v_p e \approx 6.867 \times 10^6\text{m/s} \approx 2.29\%\text{speed of light}}$$

c)

Given ...

$$d = r\left(\frac{2M}{m}\right)^{1/3}$$

$$m = 17.5M_\odot$$

$$7.4R_\odot$$

We find ...

$$\boxed{d_{rl} \approx 3.842 \times 10^{11}\text{m}}$$

$$\boxed{d_{closet} = \frac{r_p}{d} = 49 \times d_{RL}}$$